

Does a regular background bias 2D causal-set lattice-gas quantum gravity? A random-background comparison

Independent project note, v1 (2026-10-09) -- not peer reviewed

This is a short note. For full validation results, every pitfall found and fixed along the way, and commit-level citations for every number below, see `full_report.md` in this same folder.

The open question

Cunningham & Surya's dimensionally-restricted "lattice gas" [arXiv:1908.11647] makes 2D causal-set Monte Carlo cheap enough to run by placing causal-set elements on a fixed *regular* lattice embedded in a cylinder spacetime, rather than sampling generic partial orders directly. It reproduces the generic model's phase transition between a manifold-like phase and a layered, highly-ordered phase at much lower computational cost. In their Section 6, C&S note explicitly that this regular background is a modeling choice, not a physical requirement, and that checking whether it biases the result -- as opposed to using "a more realistic model of discreteness" -- is an open question they raise but do not pursue.

The answer

Yes, the location of the transition depends on the background being regular -- the effect is real, significant, and shrinks with system size, but does not vanish at the two sizes tested; the qualitative physics (phase structure, hot-phase statistics) does not depend on the background.

We built a random-background variant of the lattice gas: identical region, filling representation, Monte Carlo moves, and action, with the only change being that the m lattice sites are replaced by a fixed (quenched) uniform-random sprinkling of continuous positions. An explicit equivalence test confirms this reduces exactly to the regular lattice's own causal structure when the random positions are placed back on the integer grid, so "random background" and "regular lattice" differ only in that one respect.

What we measured

Using a Wang-Landau / multicanonical sampler (built to replace an earlier parallel-tempering scan whose results turned out to be mostly right-censored -- see `full_report.md` Section 6), we located the critical coupling `beta_c` at two system sizes, $n=30$ and $n=50$, for both backgrounds, with 5 independent realizations each:

n	Random background <code>beta_c</code>	Regular lattice <code>beta_c</code>	Gap	z	Relative shift
30	9.723 +/- 0.026 (SEM)	11.095 +/- 0.005 (SEM)	1.372	51.8	12.4%
50	5.959 +/- 0.033 (SEM)	6.419 +/- 0.005 (SEM)	0.460	13.67	7.2%

At both sizes the regular lattice's located `beta_c` is significantly higher than the random background's -- not explainable by run-to-run noise on either side ($z=51.8$ and $z=13.67$ are both far beyond any conventional significance threshold). The regular-lattice value at $n=30$ is also cross-checked against this project's independent earlier measurement of the same model (within about 2 standard deviations), and the random background's result is robust to the exact pipeline used to produce it (two different Wang-Landau sampling methods agree to within 0.008 on two directly re-checked seeds).

The gap shrinks with size, consistent with (not proven by) $1/n$ scaling. The relative shift times n is close to constant across the two sizes we tested ($12.4\% \times 30 = 371$ vs. $7.2\% \times 50 = 358$, a two-point power-law exponent of about 1.07), suggesting the absolute gap falls off at roughly the same rate as `beta_c` itself. Taken at face value, this predicts a relative shift of about 5.6% at $n=64$ -- the largest size reachable with this project's current code without a bitset- engine rewrite (every model here caps out at $N, n \leq 64$) -- a concrete number anyone can check against a direct run.

We checked and ruled out the simplest mechanism. A natural guess is that the random background simply reaches a deeper, more-ordered cold-phase action floor, which would mechanically pull `beta_c` down. Directly comparing the lowest action reached and the hot-phase mean action between the two backgrounds shows no statistically significant difference -- this hypothesis does not hold. The real mechanism (presumably something about the shape of the density of states across the explored range, not its extremes) is still open.

What does *not* change: the hot phase ($\text{beta}=0$ fillings) is close to background-independent at both sizes -- every measured observable (Myrheim-Meyer dimension, ordering fraction, height, occupied row count) lands within about one standard deviation of the regular lattice's own value. The cold phase differs more at $n=30$ but that gap has nearly closed by $n=50$ (ordering fraction 0.661 vs. 0.660, MM dimension 3.60 vs. 3.60). Neither background, at either size, shows a resolved double-peaked action histogram at its own critical coupling -- so this project does not confirm a first-order transition on either background, and the random-background result above is about *where* the transition sits, not about resolving its order.

Caveats

- **Only two system sizes.** The $1/n$ -consistent shrinkage above is a two-point observation, not a fitted law; a third size ($n=64$ is the concrete next test) would meaningfully sharpen this.
- **Mixed sampler methods at $n=50$.** Two Wang-Landau bugs were found and fixed partway through the $n=50$ runs (an unphysical histogram cliff from a flat-fill widening step, and a genuinely kinetically-isolated bin that no amount of reweighting could repair -- both described in full in `full_report.md` Section 6). The $n=50$ dataset mixes 5 seeds run under the original method with 5 run under the fixed method; a direct 2-seed consistency check found no material difference (0.002 and 0.008), but a fully single-method $n=50$ dataset was not produced.
- **Mechanism still open.** We ruled out the simplest explanation for *why* the backgrounds differ, not the real cause.
- **2D only.** C&S's own paper, and the underlying causal-set action, both extend to higher dimensions; this project has not run a 3D or 4D version of either model.

Where to go next

`full_report.md` in this same folder has the complete validation story (the order model checked against the published finite-size- scaling formula, the lattice gas checked against C&S's own quoted numbers), every methodological pitfall found and fixed along the way, and exact commit-level citations for every number in this note. The code, tests, and scripts to reproduce any of it are in the same repository; see `README.md` at the repository root.